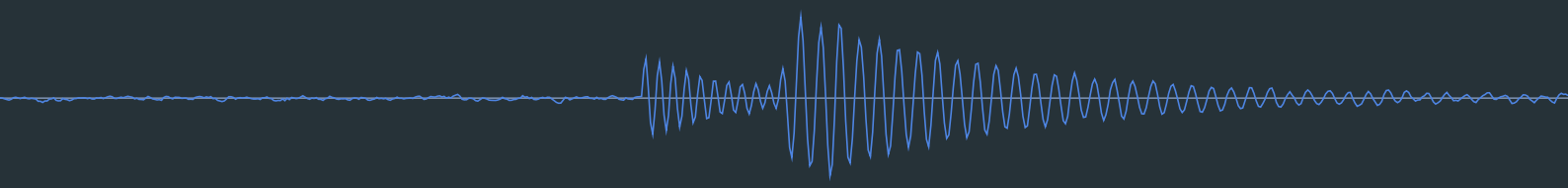
RS3D

Raspberry Shake 3D



User’s Guide

Status, live view, data download and analysis, and the helicorder for your Raspberry Shake 3D

Edition 1.0 • October 2026 • for rs3d-app 0.1 and later

Companion to the RS3D Theory and Algorithms Manual. Every screenshot was captured from the running application.

About this guide

RS3D is a desktop application for the **Raspberry Shake 3D** seismograph. It runs on your laptop, talks to the Shake over an Ethernet cable, and gives you four things: a health check of the Shake and its connection, a live view of the three ground-motion channels, a tool to download, inspect and export recorded data, and a 24-hour *helicorder* with a detail view for looking closely at any moment.

This guide teaches you to use it, from the first start to troubleshooting. It is task-oriented: each chapter starts from something you want to do. If you want to know *why* a plot looks the way it does, or how a number is computed, the companion *Theory and Algorithms Manual* explains it; the two manuals refer to each other by chapter.

Who this guide is for. Anyone who owns a Raspberry Shake 3D and wants to look at its data on a laptop: hobbyists, students, teachers and researchers. No programming is required to use the application; the optional Python script it can generate is for people who want to go further.

How the screenshots were made. Every screenshot was captured from the running application by a script. To avoid publishing anyone's real station, the examples use a *synthetic* recording of a fictitious station called `DEMO1`: one day of noise containing a local earthquake at 03:12 UTC, a distant earthquake at 14:40, two noise bursts and two data gaps. Your own data will look different, usually quieter and without so many events.

Conventions used in this guide

Bold sans-serif is text you see on screen: a button, a page or a label. `Monospace` is something you type or a file name. A Key is a keyboard key. Numbered red circles in a screenshot refer to the numbered list next to it.

Where alerts are

Earthquake *alerts* (a notification when the ground starts to shake) are planned and are not part of this edition. Their theory is in Chapter 10 of the Theory Manual; a chapter on using them will be added when the feature ships.

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1 Introduction

1.1 What RS3D is

A Raspberry Shake 3D is a small seismograph: three geophones (one vertical, two horizontal) and a digitiser in one box, recording 100 samples per second on each of its three channels. It records data on its own SD card and can also stream it live over the network. RS3D is the program on your laptop that makes use of both.

1.2 What you can do with it

Check the Shake The **Status** page tests the cable, the live data port, the Shake's clock and storage, and tells you in plain language what is wrong and how to fix it (Chapter 4).

Watch live The **Live** page draws the three channels as the ground moves (Chapter 5).

Work with data The **Data** page downloads recorded days from the Shake, lets you inspect any time range as waveforms, a spectrogram and a noise spectrum, shows how complete the data are, and exports to common formats (Chapter 6). It also opens miniSEED files from any other station.

See the whole day The **Helicorder** page shows 24 hours at a glance. A click opens a zoomable detail view with spectra and a spectrogram, and a one-click picture of the result (Chapter 7).

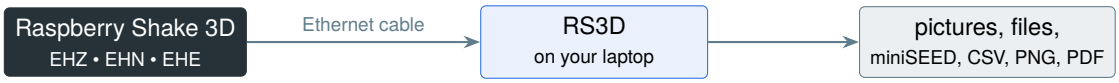


Figure 1.1: The Shake is connected to the laptop with a network cable; RS3D reads it live and copies its stored data.

1.3 What you need

- A Raspberry Shake 3D with its power supply and a network cable.
- A laptop or desktop computer with an Ethernet port (or an adapter). The application was developed and tested on Ubuntu Linux; code paths for Windows and macOS exist, but the authors have not verified them.
- Python 3.11 or newer, and the `uv` tool that installs the application's libraries (Chapter 2).
- A web browser. The application's pages open in your browser; the application itself runs on your own computer, and nothing is sent to the Internet unless you ask for the instrument response (Section 6.4).

1.4 How it relates to other Raspberry Shake software

Raspberry Shake provides its own tools; RS3D complements them rather than replacing them.

Raspberry Shake web page every Shake serves a configuration page in a browser at its own address.

rsudp a program from Raspberry Shake that receives the stream on a computer, plots it and raises alerts (Raspberry Shake S.A. 2026b).

DataView Raspberry Shake's web application for looking at data from many stations (Raspberry Shake S.A. 2026a).

RS3D focuses on *your own Shake on your own laptop*: it works without Internet, keeps the data on your disk, and adds the data-quality, spectral and helicorder tools described in this guide.

1.5 The application at a glance

When you start RS3D a sidebar on the left lists your devices and the five pages. The header shows the page you are on.

Hover the mouse over any (?) icon for a one-line explanation, and click it for the full text.

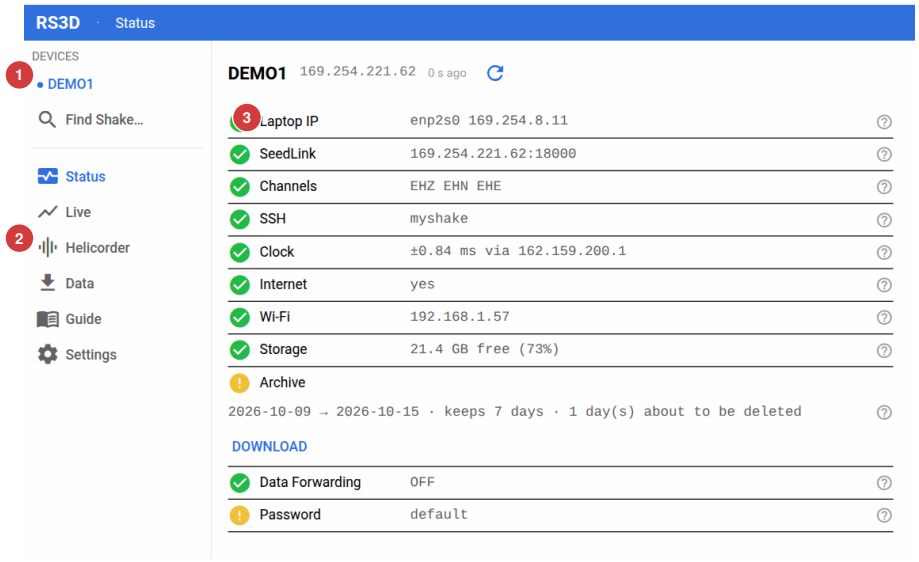


Figure 1.2: The window layout. (1) Devices: the Shake or Shakes you have added, and **Find Shake...**. (2) The pages: **Status**, **Live**, **Helicorder**, **Data**, **Guide**, **Settings**. (3) The page itself.

2 Installation and first run

2.1 Install

RS3D is distributed as source code. It needs Python 3.11 or newer and uv, a tool that creates an isolated environment with exactly the libraries the application needs.

```
cd rs3d-app
uv sync
```

`uv sync` reads the project's lock file and installs the libraries (ObsPy, SciPy, NiceGUI and others) into a folder named `.venv`. It does not touch the rest of your system. It needs the Internet once, the first time.

2.2 Start the application

```
uv run rs3d-app
```

Your browser opens on the application. The first time (no Shake added yet) you land on **Settings**; afterwards you land on **Status**. To stop the application press Ctrl+C in the terminal where you started it.

Choosing the port

> Without options the application listens on port 8765, or on any free port if that one is taken, and prints the address. Use `uv run rs3d-app --port 9000` to choose one, and `uv run rs3d-app --no-browser` to start it without opening a window (for example on a second computer).

2.3 Where your data is kept

Everything the application stores lives in one folder, shown on the **Settings** page next to **Data folder**.

System	Default folder
Linux	`~/.local/share/rs3d`
macOS	`~/Library/Application Support/rs3d`
Windows	`C:\Users\<you>\AppData\Local\rs3d`

To use a different folder, for example on a larger disk, set the environment variable `RS3D_DATA_DIR` before starting:

```
RS3D_DATA_DIR=/mnt/seismic/rs3d uv run rs3d-app
```

Chapter 8 explains what is inside the folder and how to back it up.

2.4 Updating and uninstalling

To update, replace the program folder with the newer one and run `uv sync` again; your data folder is separate and is kept. To remove the application, delete the program folder and, if you also want to delete your data, the data folder.

The data folder is your recordings

> Deleting the data folder deletes every day file you downloaded. The Shake itself keeps only the most recent days (7 by default), so data that exist only on your laptop cannot be fetched again.

3 Connecting a Shake

3.1 Cabling and power

1. Place the Shake on a solid, level floor, ideally on the ground floor, away from doors, fans and footsteps.

2. Connect its Ethernet port directly to your laptop's network port.

3. Power the Shake with its own supply and wait **three to five minutes** for it to start.

Without a router (no DHCP server) both ends give themselves a *link-local* address in the range `169.254.x.x`. That is normal and is what RS3D expects.

3.2 Give the laptop a 169.254.x.x address

The laptop needs an address in the same range on the wired port. Many systems do this by themselves when the cable is plugged in. If they do not, the **Status** page shows **Laptop IP** in red with a **Set IP** button (Chapter 4). Pressing it shows the exact commands, asks your system for permission and applies them.

3.3 Find the Shake

Press **Find Shake…** in the sidebar. RS3D looks for Shakes on the cable (it tries the names `rs.local`, `rs-2.local` and so on, and the addresses your computer has recently seen) and confirms each candidate by asking it for its station name. Every Shake it finds is listed with its station code and address; press **Add** to add it.

If nothing is found, check the cable, wait a little longer for the Shake to start, and see Chapter 9; you can always add the Shake by hand.

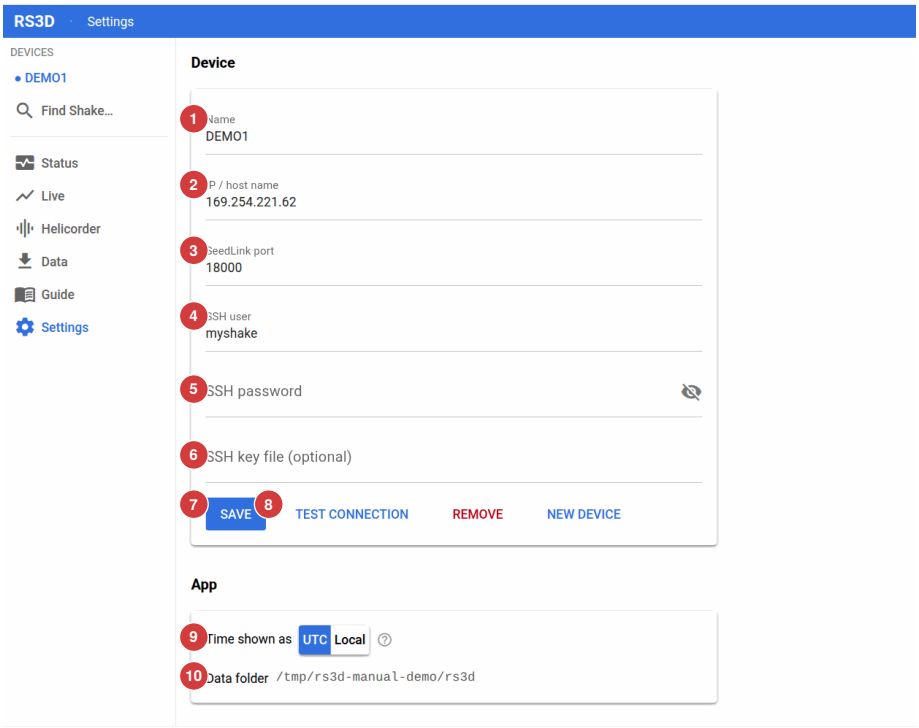


Figure 3.1: The **Settings** page. (1)–(6) The device fields, described below. (7) **Save**. (8) **Test connection**. (9) Time zone of the displayed times. (10) The data folder.

3.4 Add a Shake by hand

Open **Settings**. The **Device** card has these fields.

1. **Name**: what you want to call it in the sidebar (default: the station code).
2. **IP / host name**: the Shake's address, for example `169.254.221.62` or `rs.local`.
3. **SeedLink port**: the live-data port; 18000 unless you changed it on the Shake.
4. **SSH user**: the account RS3D uses to read the Shake's clock, storage and archive; `myshake` on a standard Shake.
5. **SSH password**: that account's password. It is stored in your operating system's key store (not in a file) and never leaves your computer.
6. **SSH key file (optional)**: a private key, if you prefer key login to a password.

Press **Test connection**; **Connection OK** means the login works. Then press **Save**.

Change the default password

> A new Raspberry Shake uses a published default password. The **Password** check on the Status page warns while it is unchanged. To change it, log in to the Shake over SSH and run `passwd`, then enter the new password in **Settings**. RS3D only *reads* from the Shake and never changes anything on it.

3.5 More than one Shake

Every Shake you add appears in the sidebar under **Devices**; click a name to switch. **New device** on the **Settings** page starts an empty form, and **Remove** deletes the selected device (its downloaded data stay on disk).

4 The Status page

The **Status** page answers one question: *is everything in order?* It runs eleven read-only tests on your laptop's link to the Shake and on the Shake itself, and shows one row for each. It refreshes by itself every 30 seconds; the small text next to the Shake's name shows how long ago the last check ran, and the round arrow runs it again.

4.1 Reading the icons

- **green**: the check passed.
- **yellow**: it works, but you should look at it (for example the Shake still uses its default password).
- **red**: something is broken and the application cannot rely on it.
- **grey**: not tested, because a check it depends on failed (no cable, no SSH login).

Fix the first red row from the top: the checks below it are usually grey because of it.

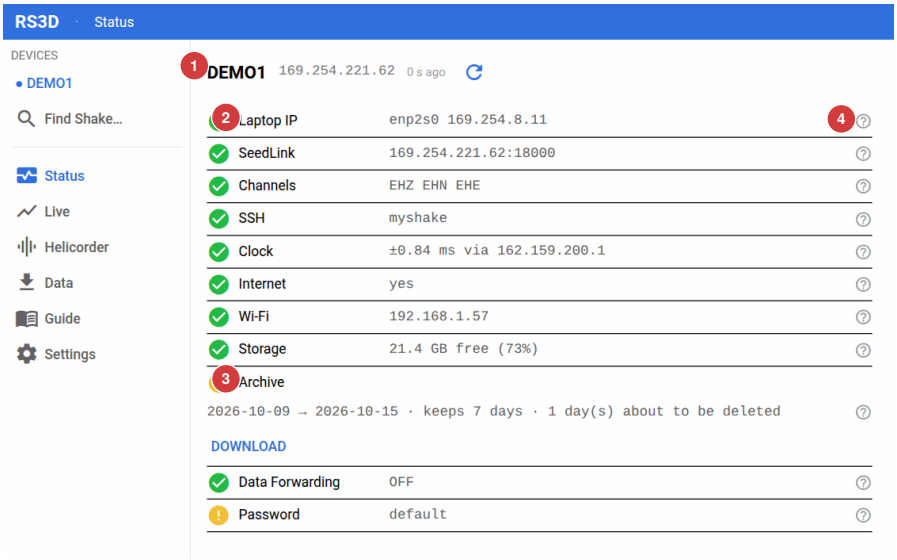


Figure 4.1: A healthy Shake. (1) The Shake's name and address, and the refresh button. (2) One row per check: icon, name and the measured value. (3) A yellow icon is a warning. (4) The (**?**) icon explains each check.

Table 4.1: What each check tests and when it complains.

Check	What it tests	Warning or error when
Laptop IP	the laptop has a 169.254.x.x address on an active wired port	none: red, with **Set IP**
SeedLink	the Shake's live-data port (TCP 18000) answers	unreachable: red
Channels	the Shake serves `EHZ`, `EHN`, `EHE`	a channel is missing: yellow
SSH	the application can log in as the SSH user	wrong password, unreachable, new host key: red
Clock	offset between the Shake's clock and its time source	above 100 ms yellow; above 1000 ms or no source red
Internet	the Shake can reach the Internet (needed for its clock)	no: yellow
Wi-Fi	the Shake's Wi-Fi address	two addresses on the Wi-Fi port: yellow
Storage	free space on the Shake's SD card	under 10 % free: yellow
Archive	the days stored on the Shake and its retention	none stored, or days about to be deleted and not downloaded: yellow
Data Forwarding	whether the Shake shares its data publicly	`ON`: yellow (data and location are public)
Password	whether the default password was changed	still default: yellow

4.2 The eleven checks

Clock. Earthquake timing depends on the Shake's clock, which is kept right by the network time protocol. The value shows the offset in milliseconds and the time source. A Shake with no source drifts, and after a power cut restarts from an old time. With the laptop on a direct cable and no router, a Shake normally obtains its time over its Wi-Fi Internet connection, which is why the **Internet** and **Wi-Fi** checks come next.

Archive. The Shake writes one file per channel per day and deletes days older than its retention (7 days on a standard setup). When some days are about to be deleted and are not on your laptop, the row turns yellow and shows a **Download** button that opens the **Data** page. Download at least once a week.

Data Forwarding. When this is `ON` and the Shake has Internet access, its data and approximate location are public on Raspberry Shake's *ShakeNet*. If you did not intend that, turn it off in the Shake's own web page (*Data Sharing*).

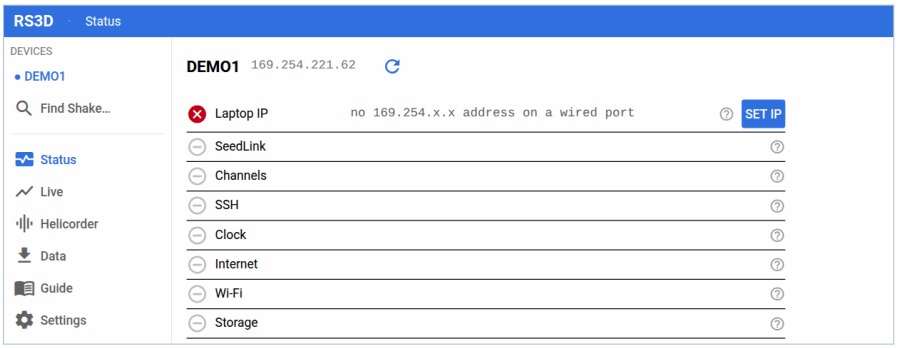


Figure 4.2: The **Laptop IP** check failing. All other checks are grey until it is fixed.

4.3 When the laptop address is missing

If the **Laptop IP** row is red, the cable may be unplugged, the Shake may still be starting, or your system did not assign a link-local address. The row offers **Set IP**.

Set IP opens a dialog where you choose the wired port and see the *exact commands* that will run. Press **Run**; your system asks for permission, and the page checks again. On Linux the application asks NetworkManager to enable a link-local address on that connection; on Windows and macOS it switches the adapter to automatic (DHCP) addressing, and the system then assigns the 169.254.x.x address itself.

4.4 Fixing the other rows

Every row has a (**?**) icon; its text states the cause and the fix for the problem it found. The most common ones are in Chapter 9.

5 The Live page

The **Live** page draws the three channels as they arrive from the Shake. It needs only the cable and the live-data port, not the SSH login, and keeps working while the Status page shows warnings.

5.1 The controls

Window how many seconds are shown: **30 s**, **60 s**, **5 min** or **10 min**. The page keeps the last ten minutes of data; a longer window than the data collected so far is simply partly empty.

Filter a band-pass filter applied to what you see, between the two frequencies in the boxes (1 and 10 Hz at first). It hides the slow drift and the high-frequency noise so that the earthquake stands out. Switch it

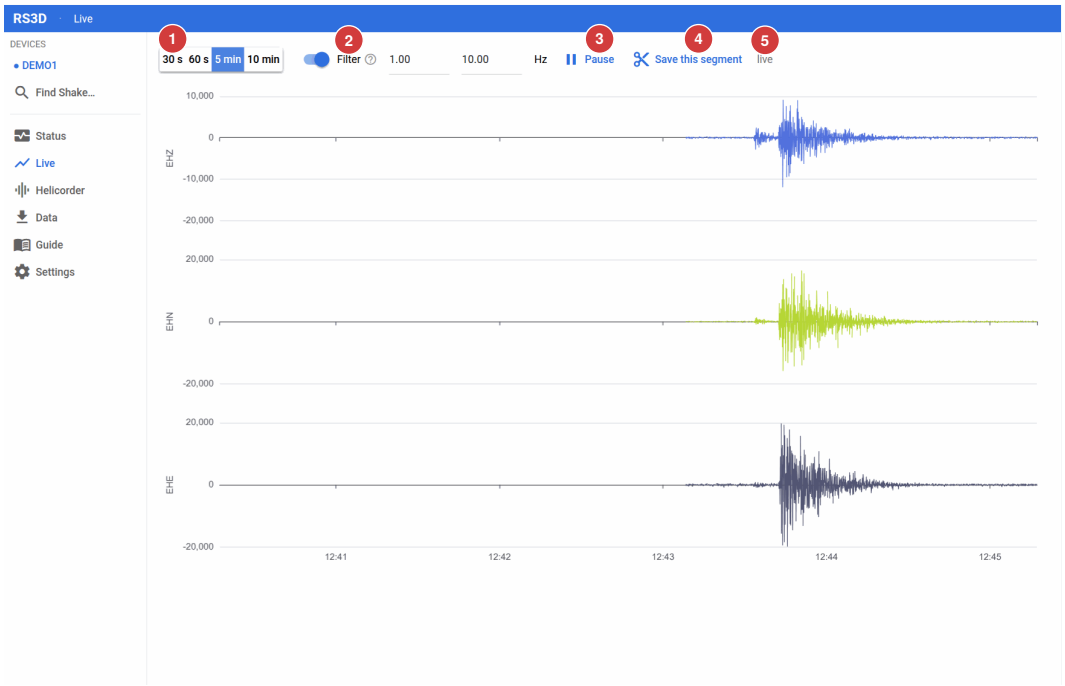


Figure 5.1: The Live page showing the demonstration earthquake arriving on all three channels. (1) Time window. (2) Filter switch and band. (3) Pause. (4) Save this segment. (5) Connection state.

off to see the raw signal, or change the numbers. The filter changes only the display (Chapter 4 of the Theory Manual).

Pause freezes the picture so that you can look at it. Data keep arriving in the background; **Resume** jumps to the present.

Save this segment opens the **Data** page on exactly the interval that is displayed, so you can inspect or export it (Chapter 6). The data must exist in the Shake's archive for it to be downloaded.

Connection state at the right of the toolbar, see below.

5.2 The channels

The three plots, from top to bottom, are `EHZ` (vertical), `EHN` (north–south) and `EHE` (east–west). Each has its own amplitude axis in *counts*, the raw numbers from the Shake's converter. The time axis is in UTC or in your local time, as chosen in **Settings**.

5.3 Connection states

Shown	Meaning
connecting…	the application is opening the connection
live	data are arriving
waiting for data…	the connection is open but no data has arrived for five seconds
reconnecting…(reason)	the link was lost; the reason (for example *connection refused*) is shown, and the application tries again after 2, 4, 8, 16 and then every 30 seconds
stopped	the stream was stopped

You do not need to do anything: when the Shake comes back, the page resumes by itself.

A delay of a few seconds is normal

> The Shake sends data in short blocks; the picture is typically a few seconds behind the ground. This is a property of the instrument's data blocks, not of the application (Theory Manual, Section 11.5).

6 The Data page

The **Data** page is where recorded data are downloaded, inspected and exported. The workflow is always the same: **pick a station, pick a time range, choose how to process it, press the big button, then look and export.**

6.1 Choose a station

The **Station** list (1) contains your Shake, and every other station whose data are on this laptop (for example from imported files). A station name has three parts separated by dots: network, station and location, for example `AM.R77A7.00`.

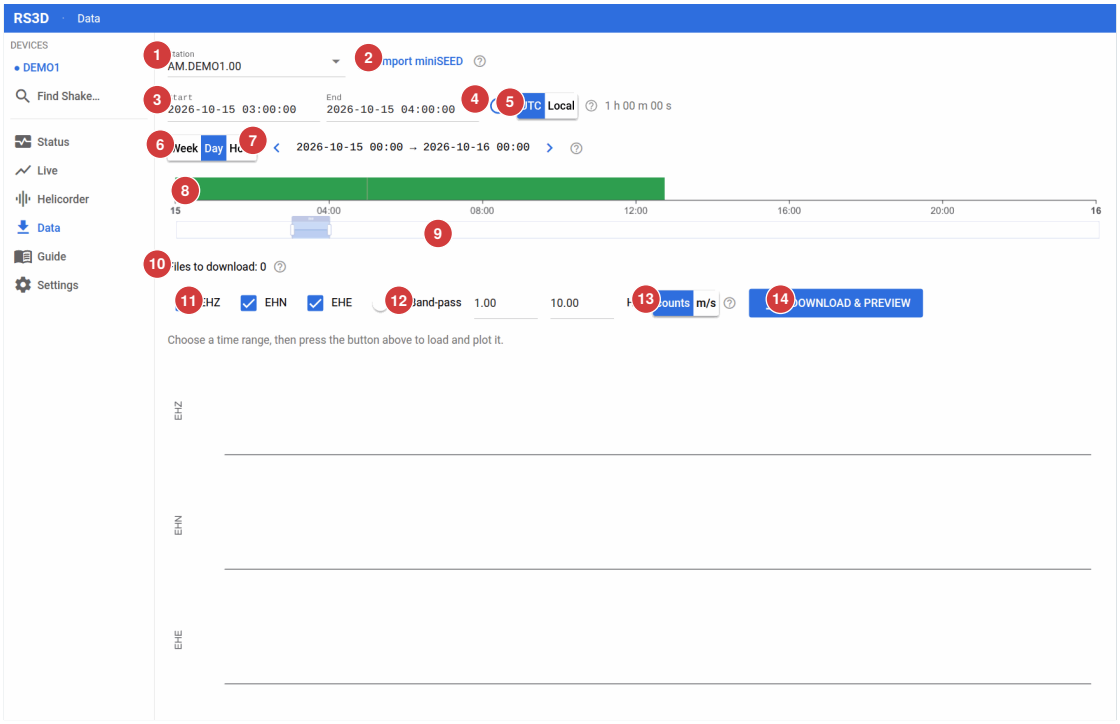


Figure 6.1: The Data page before a preview. (1) Station. (2) Import a miniSEED file. (3) Start and end. (4) Presets. (5) UTC or local time. (6) Visible span of the coverage bar. (7) Move the visible span. (8) Coverage bar. (9) Selection handle. (10) Files that will be downloaded. (11) Channels. (12) Band-pass. (13) Units. (14) Download and preview.

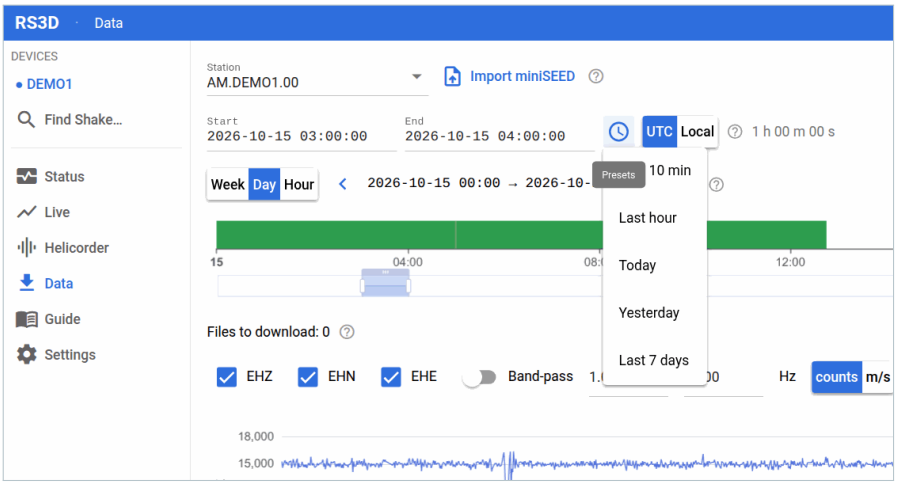


Figure 6.2: The presets menu: **Last 10 min**, **Last hour**, **Today**, **Yesterday** and **Last 7 days**.

6.2 Choose a time range

Typing. The **Start** and **End** fields (3) take a date and time written as year-month-day hour:minute:second, for example `2026-10-15 03:00:00`. Press Enter or click away to apply. A malformed value shows a hint with the expected form. The text next to the toggle shows the length of the range.

Presets. The clock button (4) opens a menu of ready-made ranges.

UTC or local. Seismic data are always stored and exported in UTC. The toggle (5) only changes the time zone in which the fields and axes are *displayed and typed*; **Today** and **Yesterday** follow it. The code and the files you export always use UTC.

6.3 The coverage bar

The coloured bar (8) shows what exists for the visible span.

- **green**: the data are on this laptop
- **grey**: the data exist only on the Shake and have not been downloaded
- **red**: no data anywhere (a gap, or time that has not happened yet appears blank)

The strip below it (9) is the selection. Drag in it to select a range, drag its handles to change the start or end, or drag the middle to move it; the **Start** and **End** fields follow. **Week**, **Day** and **Hour** (6) change how much time the bar shows, and the arrows (7) move to the previous or next span.



Figure 6.3: The coverage bar for 15 October: green where the demo data are on the laptop, and the selection (light blue) about 03:00–04:00.

Files to download. The number next to (10) is how many day files will be copied from the Shake when you press the button. Today's file is always copied again, because the Shake keeps writing to it.

6.4 Choose channels, filter and units

Channels (11). Tick the channels to load: `EHZ` (vertical), `EHN` (north–south), `EHE` (east–west).

Band-pass (12). Switch it on to keep only the frequencies between the two boxes (1 and 10 Hz at first) and remove drift and high-frequency noise. The band you enter is remembered for next time. The filter is zero-phase, so it does not delay the waveform, but it smears sharp onsets slightly backwards in time; do not use it to read exact arrival times (Theory Manual, Chapter 4).

Units (13). **counts** are the raw numbers of the converter. **m/s** converts them to ground velocity using the instrument response. The first time you choose **m/s** for a station the application downloads the response once from the Raspberry Shake service (the laptop needs Internet that one time) and keeps it, after which the switch works offline. Without it and without Internet you get an error message instead of a chart.

6.5 Download and preview

Press the large button (14). For your own Shake it reads **Download & preview**: it first copies the day files you do not yet have, showing progress (*Downloading 2/6: EHZ 2026-10-14*), then draws the data. For a station that is not a connected Shake (for example an imported file) the button reads **Preview** and only reads local files. If the Shake is switched off the download is skipped, an explanation is shown and whatever is already on the laptop is displayed.



Figure 6.4: The waveform preview of one hour (03:00–04:00): three channels sharing one time axis. The short spike near 03:12 is the local earthquake; the amplitude axes are in raw counts.

6.6 Reading the preview

Waveforms. One plot per selected channel, with the time axis in the zone you chose. Hover for the value at a point. The counts do not oscillate around zero because they sit on the instrument's offset (Theory Manual, Section 3.4); switching the band-pass on removes it.

Data health. Just under the export buttons, one line says how complete the selected range is and how many gaps it has, for example *100.00 % complete · no gaps*. If there are gaps their start, end and length are listed in the time zone you chose (Theory Manual, Chapter 8).

Spectrogram. A picture of *which frequencies were present when*. Time runs left to right, frequency (0 to 50 Hz) bottom to top, colour is power in decibels (red strong, blue weak). Use the channel selector next to the title to choose `EHZ`, `EHN` or `EHE`. An earthquake is a vertical streak; a steady horizontal line is a constant interfering frequency, here a 30 Hz line that is part of the demonstration data. Blank columns are gaps.

Noise spectrum. The median level of each frequency over the whole selection, with a band showing the 10th to 90th percentile. It is the quickest way to compare a quiet period with a busy one: select night, preview, select midday, preview again. It is in raw units and is meant for *comparing your own selections*, not for comparing with published noise models (Theory Manual, Chapter 7).

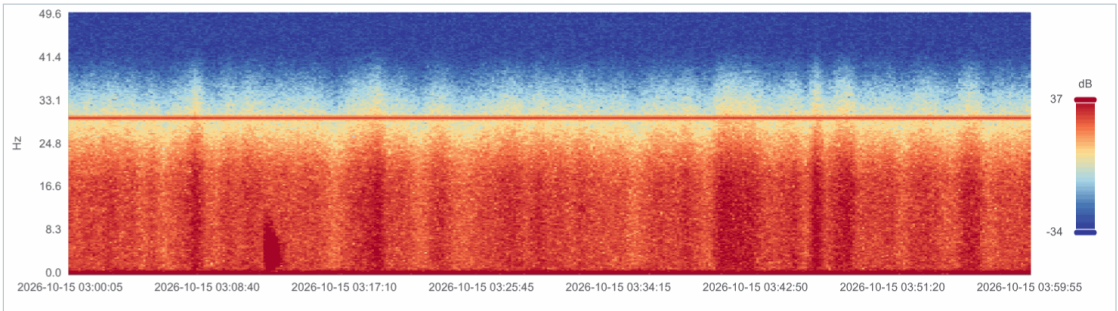


Figure 6.5: The spectrogram of the same hour for `EHZ`. The horizontal red line near 30 Hz is a steady interference; the dark patch near 03:12 is the earthquake.

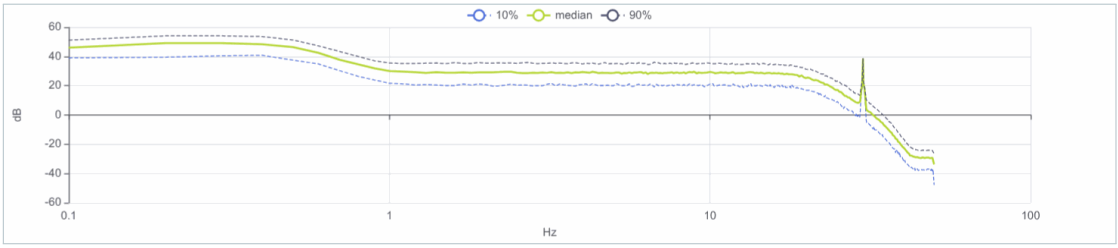


Figure 6.6: The noise spectrum: median (green) and 10th/90th percentile (dashed). The spike at 30 Hz is the interference line; the fall near 50 Hz is the Shake's anti-alias filter.

6.7 Exporting

After a preview the export buttons below the waveforms become active: **miniSEED**, **CSV**, **PNG**, **PDF** and **</> Code**. The file name is the network, the station, the start and the end time in UTC, and the extension. For example:

```
AM.DEMO1_20261015T030000_20261015T040000.mseed
```

Exports contain exactly what the preview shows: if the band-pass was on, the file is filtered; if you chose **m/s**, it is in m/s.

miniSEED — the standard seismological format, readable by ObsPy, SeisComP and most other tools. Gaps are kept.

CSV — a text table: one column of UTC times and one column per channel, one row per sample; missing samples are empty. All channels must have the same sampling rate. See Appendix B.

PNG, PDF — a picture of the waveforms for reports.

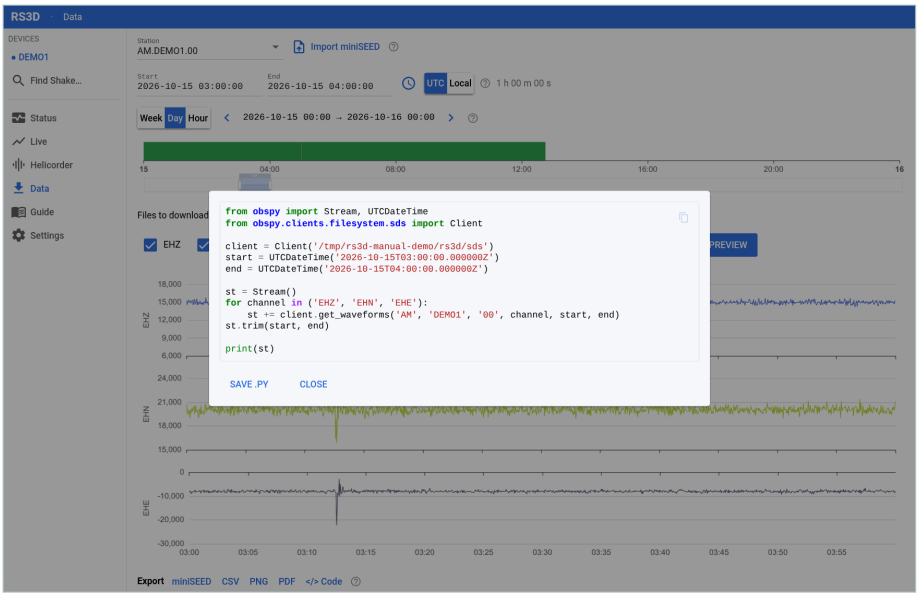


Figure 6.7: The generated Python program for the preview above.

</> Code opens a dialog with a short Python (ObsPy) program that reproduces *exactly* the processing behind the preview, and a button to save it as a `.py` file. It reads from this laptop's data folder, so it runs without the Shake. Use it when you want to go further than the application does, or to document how a number was obtained.

6.8 Importing miniSEED files

Import miniSEED (2) adds a file from anywhere, for example from another Shake, a university network or a data centre. Accepted names end in `.mseed`, `.msd`, `.miniseed` or `.ms`, up to 2 GB. The file is split into one file per channel per day in your data folder, next to the data from your Shake, and the page switches to the imported station. Importing the same file twice changes nothing; importing a file with overlapping data merges them. Imported stations then appear in the **Station** list, on the **Helicorder** page and everywhere else.

No Shake needed

Import is also the quickest way to try the application: the whole Data and Helicorder pages work on any miniSEED file.

7 The Helicorder

A *helicorder* wraps a whole day of seismic data into rows, like lines of text, so that a day fits on one screen and an event stands out as a burst on one line. The **Helicorder** page shows one channel of one day; a click on any row opens a detail view of that moment.

7.1 Opening the helicorder

Choose **Helicorder** in the sidebar. The page shows the newest day for which data exist on the laptop. The helicorder reads the data that are *already on your laptop*: for a day you have not downloaded yet, first use the **Data** page (Chapter 6); a day without data is shown as empty with a message.

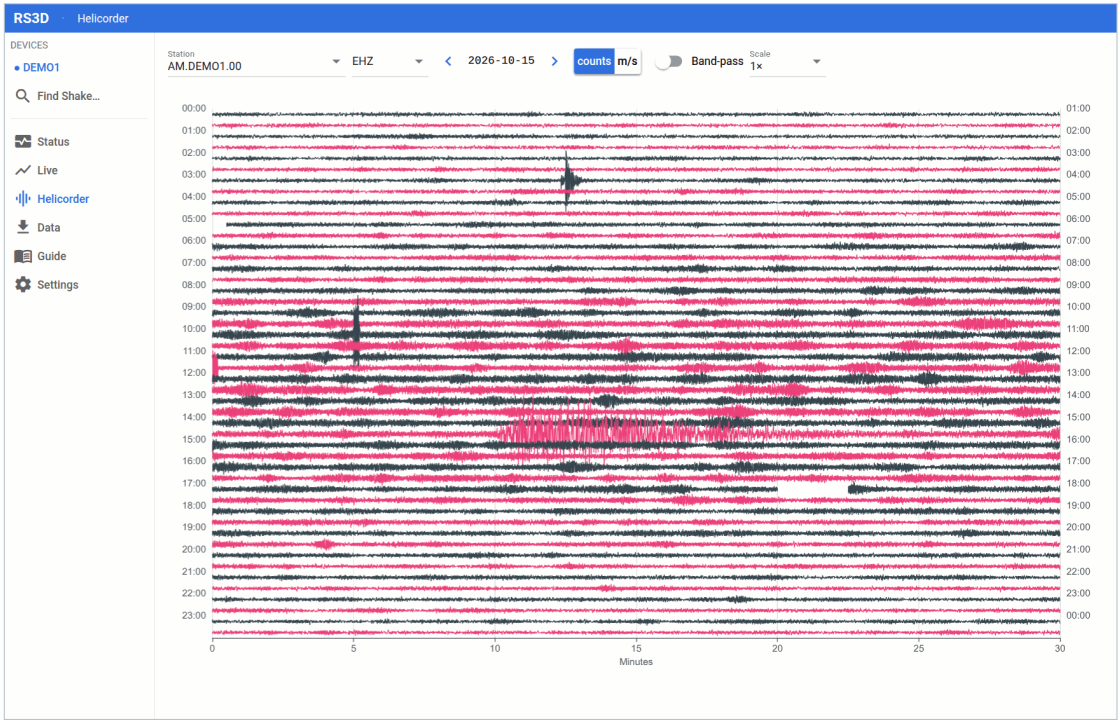


Figure 7.1: One day of the demonstration station. Each row is 30 minutes; rows alternate in colour. The local earthquake is the burst in the 03:00 row, the noise bursts are the narrow spikes around 10:00 and 11:30, and the white gap in the 17:00 rows is a data gap.

7.2 The controls

The row of controls above the plot has, from left to right:

Station, channel	the station and the channel to draw (the vertical `EHZ` first).
⟨ **date** ⟩	the day shown; the arrows move to the previous and the next day.
counts / m/s	raw counts or ground velocity (Section 6.4; **m/s** needs the instrument response).
Band-pass	switches a band-pass filter on or off; the band is the one last set on the **Data** page (1–10 Hz at first).
Scale	makes the traces taller or shorter (0.5× to 8×); see below.

7.3 Reading the day view

Time runs from left to right along a row (0 to 30 minutes, labelled *Minutes*); rows follow each other from 00:00 at the top. The labels at the **left** are UTC hours; the labels at the **right** are the same hours in your local time. A quiet row is a thin line; shaking makes it fatter; a hole is blank.

The amplitude scale. All rows share *one* scale, so rows can be compared. At 1× the ordinary signal fills about one row and only the strongest one per cent exceeds it, so a single huge event does not flatten the rest of the day. If everything looks too flat use **2**× or **4**×; if rows overlap too much use **0.5**×. Each row has its own offset removed first, so the instrument's DC offset does not push traces out of their rows (Theory Manual, Section 9.2).

7.4 Opening the detail view

Click anywhere on a row. A panel opens above the day view showing about two minutes centred on the point you clicked. (The click is accurate to about ten seconds; use the buttons below to adjust.)



Figure 7.2: The detail view after clicking the 03:00 row at 12 minutes. (1) Station, channel, time range and units. (2) Fit. (3) Box zoom. (4) Zoom out. (5) Zoom in. (6) Earlier. (7) Later. (8) Waveform, Spectra, Spectrogram. (9) Clipboard. (10) Close.

7.4.1 The toolbar

Zoom out, zoom in (4, 5)
show twice or half as much time around the same centre, between 5 seconds and one hour.

Earlier, later (6, 7) move the window by half of its length.

Fit (2) rescales the vertical axis to the data that are in view and removes any amplitude zoom. The time window is kept. To go back to the original two minutes, click the row again.

Box zoom (3) see below.

Waveform / Spectra / Spectrogram (8)
choose how the window is drawn.

Clipboard (9) opens the Waveform Clipboard (Section 7.7).

Close (10) hides the panel.

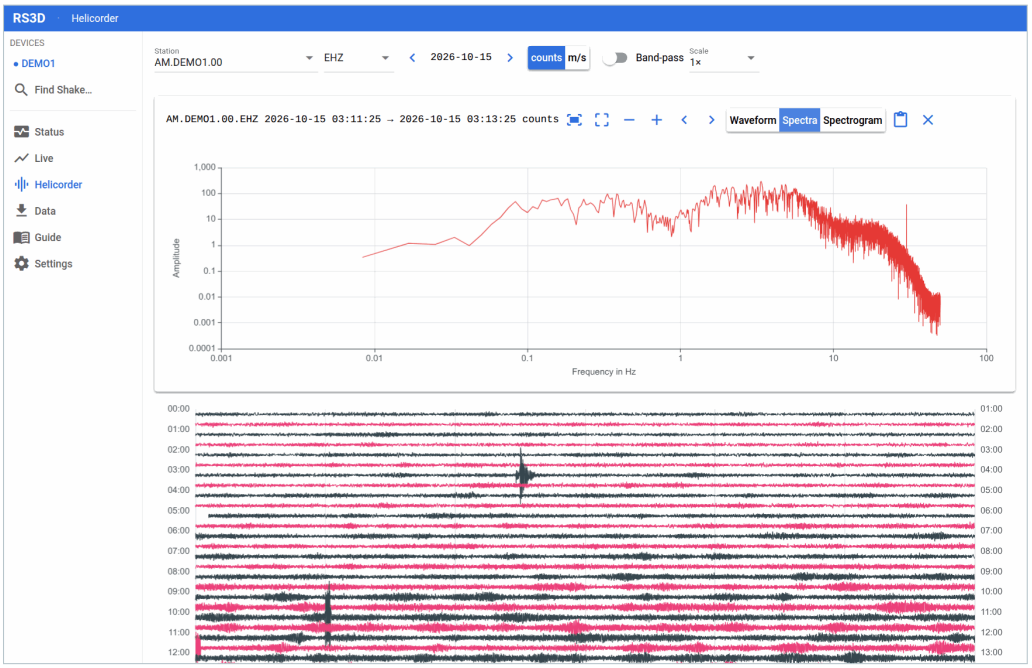


Figure 7.3: Spectra mode: amplitude spectrum of the same two minutes.

7.5 Three views of the same window

Waveform is the amplitude against time. Here the P wave arrives at 03:12:20 and the larger S wave a few seconds later.

Spectra shows how much of each frequency the window contains, on logarithmic axes: frequency in hertz horizontally, amplitude vertically. It is good for finding a steady interfering frequency and for seeing where an earthquake has its energy (Theory Manual, Chapter 5).

Spectrogram shows the frequencies against time for the window: the sudden broadband arrival of the earthquake and its fading coda stand out clearly (Theory Manual, Chapter 6).

7.6 Zooming in with a rectangle

The **Box zoom** tool lets you drag a rectangle on the waveform and zoom into exactly that part, in time *and* in amplitude.

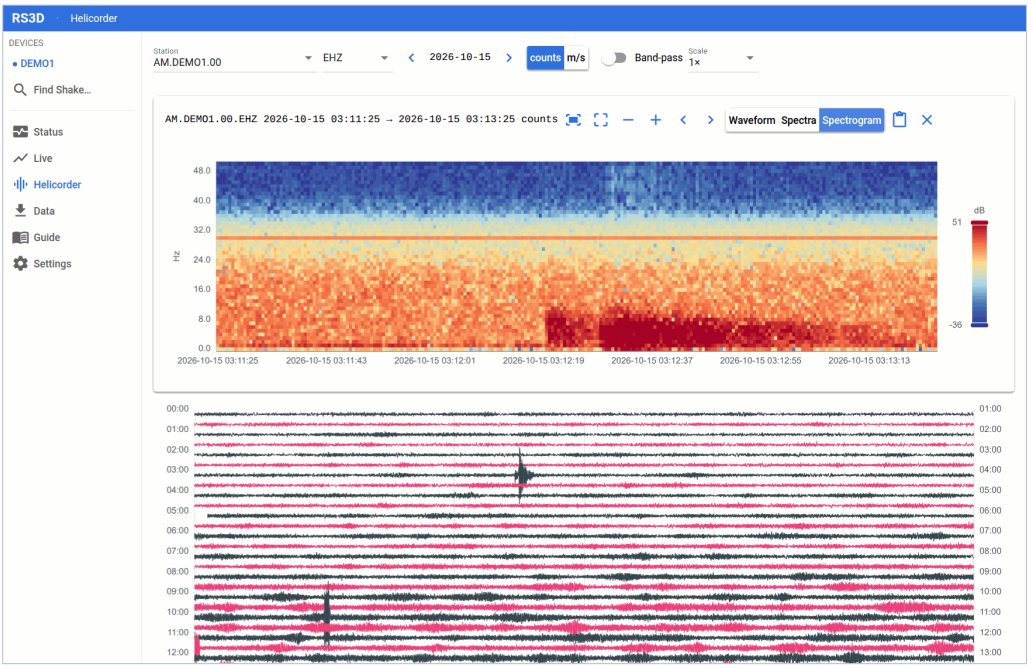


Figure 7.4: **Spectrogram** mode.

1. Press the Box zoom button (3); it turns blue.

2. Press the mouse button at one corner of the part you want, drag to the opposite corner, and release.

3. The window now shows only the time range of the rectangle, reloaded at full detail, and the vertical axis shows only its amplitude range.

Switch the button off to go back to normal mouse behaviour. The tool works in **Waveform** mode.

A rectangle narrower than five seconds is widened to five seconds around its centre. **Fit** removes the amplitude zoom; the zoom buttons change the time span as before.

7.7 The Waveform Clipboard

The clipboard button (9) opens a dialog with the three views stacked: waveform, spectrogram and spectrum, for the window that is currently on screen. The camera button at the top right saves them as **one picture** (PNG) that your browser downloads; the picture has a title with the station, channel, time range in UTC, units and filter. The cross closes the dialog.

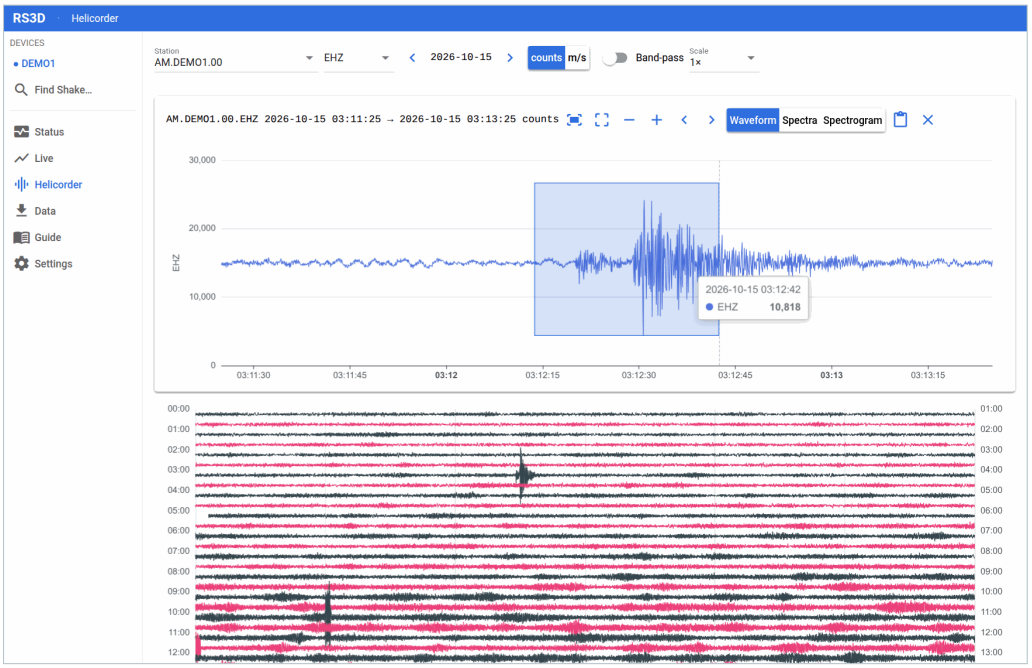


Figure 7.5: Dragging a rectangle around the earthquake onset (the blue rectangle).



Figure 7.6: After releasing: 29 seconds from 03:12:13 to 03:12:42. The P wave arrives at 03:12:20 and the stronger S wave at about 03:12:29. The vertical axis now shows only the amplitudes of the rectangle.

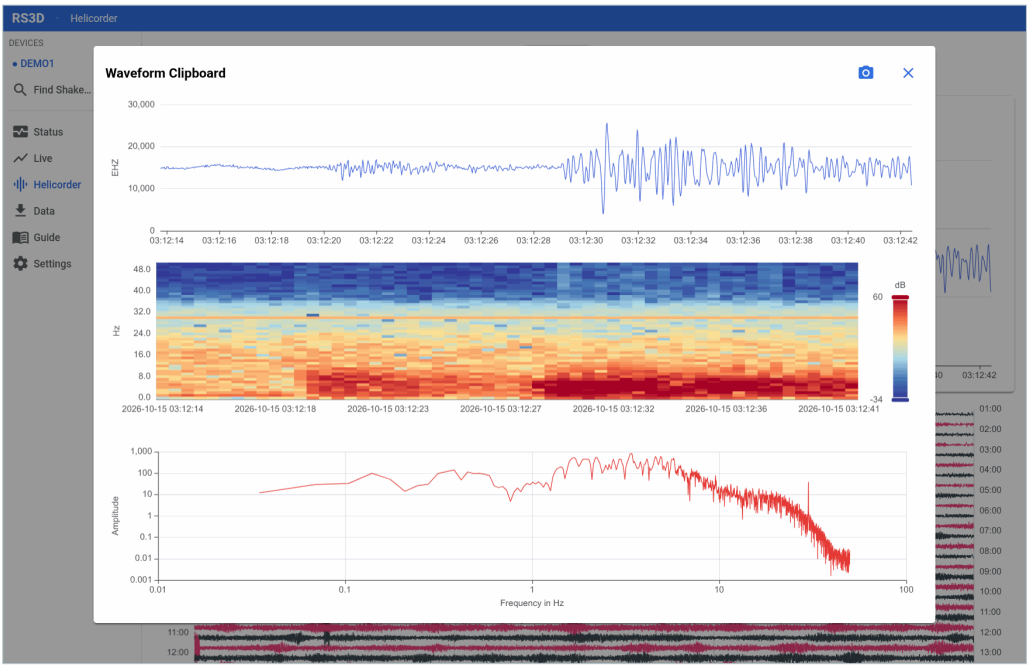


Figure 7.7: The Waveform Clipboard for a 20-second window. The camera saves the three panels as one picture.

7.8 Tips

- To find events quickly, scan the day view for fat or tall spots on a row, then click them.
- Turn the band-pass on (1–10 Hz) to see local earthquakes more clearly; turn it off to see the real signal.
- A very short event may be one burst on a single row; use **Box zoom** on its onset to read the P and S arrival times to a fraction of a second.
- Man-made noise (footsteps, a door, a vehicle) looks like a short burst without the slowly fading tail that an earthquake has.

8 Settings, preferences and storage

8.1 The Settings page

Besides the device form (Chapter 3) the **Settings** page has an **App** card with two items.

Time shown as — **UTC** or **Local**. It changes the time zone of the fields and axes on the Live, Data and Helicorder pages; stored data and exports are always UTC. The same toggle is on the Data page.

Data folder — the folder where everything is kept (Section 8.3).

8.2 What the application remembers

The application remembers, between sessions, the Shake you were using, your choice of UTC or local time, and your last choices on the data pages: the station, **counts** or **m/s**, whether the band-pass was on, and its two frequencies. These are saved the moment you change them. Passwords are not saved in any file: they go to your operating system's key store.

8.3 Inside the data folder

Item	Contents
`sds/`	your data: one miniSEED file per channel per day, in the SDS layout (Appendix B)
`response/`	instrument responses (StationXML) downloaded for the **m/s** switch
`config.toml`	the list of Shakes (name, address, ports, user; no passwords)
`settings.toml`	the preferences described above
`known_hosts`	the SSH identities of your Shakes
`logs/`	log files

Back up. Copy the whole `sds/` folder (and, if you like, `config.toml`). It contains ordinary files that other seismology software can read directly.

Move to another disk. Close the application, copy the folder, and start the application with `RS3D_DATA_DIR` set to the new place (Section 2).

Reset the stored Shake identity. If a Shake's SD card is replaced, its SSH identity changes and the **SSH** check reports it. Delete the line of that address from `known_hosts` and refresh the Status page.

Free space. A day of three channels occupies about 85 MB, so a year is about 30 GB.

The Shake keeps only a week

> The Shake deletes days older than its retention (7 days by default). Download new days regularly (the Status page warns when days are about to be deleted); your laptop's copy may then be the only one.

9 Troubleshooting

Start with the **Status** page: its red and yellow rows already say what is wrong, and the (**?**) icon next to a row explains the fix. The tables below collect the most common situations by where you meet them.

9.1 Connecting

What you see	Likely cause	What to do
Find Shake… finds nothing	the Shake is still starting; no cable; no 169.254.x.x address on the laptop	wait three to five minutes; check the cable; open **Status** and use **Set IP**; then try again, or add the Shake by hand with its address
Laptop IP red	the wired port has no link-local address	press **Set IP** (Chapter 4); check the cable and that the Shake has power
SeedLink red, *connection refused*	the Shake is not ready, or the address is wrong	wait; check the address in **Settings**; check the cable
SSH red: wrong user or password	credentials differ from the Shake's	enter the correct ones in **Settings** and press **Test connection**
SSH red: host key changed	the Shake's SD card was replaced or re-installed	delete its line from `known_hosts` in the data folder (Section 8.3) and refresh
Clock yellow or red	the Shake has no Internet or no time source	fix **Internet** and **Wi-Fi** first; give it a few minutes to synchronise
Wi-Fi yellow, two addresses	two network managers on the Shake	see the **Guide** page, "Wi-Fi: two addresses"

9.2 Live

What you see	Likely cause	What to do
reconnecting… with a reason	the link dropped; the page retries by itself	check the cable and the Shake's power; it resumes when the Shake is back
waiting for data…	connected but no data for five seconds	wait; if it persists, restart the Shake
Flat lines	the filter band is wrong, or the Shake is quiet	switch **Filter** off to see the raw signal; tap the table or stamp near the Shake to test

9.3 Data and helicorder

What you see	Likely cause	What to do
No data in this range	nothing is on the laptop for that time	check the coverage bar: grey means "on the Shake, not yet downloaded": press **Download & preview**; red means nothing was recorded
No data yet. Connect a Shake, or import a miniSEED file.	the data folder is empty	download from the Shake, or import a file
Helicorder empty for a day	that day is not on the laptop	download it on the **Data** page
No data in this window (detail view)	you clicked a row or time with no samples	click another row, or use **Earlier**/**Later**
Error when choosing **m/s**	no instrument response and no Internet	connect the laptop to the Internet once so the response can be downloaded and cached
Preview is slow	a very long range, or **m/s** on a whole day	select a shorter range, or use the helicorder for overviews
Gaps in the data	the Shake was off, or the cable was unplugged	see **Data health** for exactly when; gaps cannot be filled
Spectrogram blank in places	the window touches a gap	expected: such columns are left blank on purpose

9.4 Starting the application

What you see	Likely cause	What to do
address already in use	you passed `--port` and another program uses it	start without `--port` (a free port is chosen) or pick another number
The browser does not open	started with `--no-browser`, or no default browser	open the address printed in the terminal by hand
command not found: uv	`uv` is not installed	install it from `https://docs.astral.sh/uv/`
Page does not update after upgrading	an old browser tab is open	reload the page (Ctrl+R); restart the application

9.5 Still stuck?

Look in the `logs/` folder of the data folder, and read the **Guide** page of the application. When reporting a problem, include the text of any error message, the page you were on, and what the **Status** page shows.

A Quick reference

A.1 Pages

Page	Purpose
Status	eleven read-only checks of the link and of the Shake; **Set IP**; refreshes every 30 s
Live	live traces; window 30 s–10 min; filter; pause; save the segment
Helicorder	one day in 30-minute rows; click for the detail view; box zoom; clipboard picture
Data	station, time range, coverage bar, download, preview, health, spectrogram, noise spectrum, exports, Python code
Guide	the built-in short guide and the explanations behind the (?) icons
Settings	add and edit Shakes; time zone; data folder

A.2 Mouse and keyboard

Action	Effect
Click a helicorder row	opens the detail view at that time
Drag in the coverage strip	selects a time range (Data page)
Drag a rectangle (Box zoom on)	zooms the detail waveform to the rectangle
Hover over a plot	shows the value at that point
Hover over (?)	one-line explanation; click for the full text
Enter in a time field	applies the typed time
Ctrl+C in the terminal	stops the application

A.3 Command line and environment

Item	Effect
`uv run rs3d-app`	start the application and open the browser
`--port N`	use port N (default 8765, or a free port if it is taken)
`--no-browser`	do not open a browser window
`RS3D_DATA_DIR`	use this folder instead of the default data folder

A.4 Defaults worth remembering

Setting	Default
Live band-pass	1–10 Hz, on
Live window	60 s
Detail window (helicorder)	2 minutes
Helicorder row length	30 minutes
Shake archive retention	7 days
SeedLink port	18000
Channels	`EHZ` vertical, `EHN` north–south, `EHE` east–west
Sampling rate	100 samples per second

B File formats

B.1 miniSEED and the SDS folder layout

Data are stored as miniSEED, the standard seismological exchange format, one file per channel per day, in the *SeisComP data structure* (SDS):

```
sds/YEAR/NET/STA/CHAN.D/NET.STA.LOC.CHAN.D.YEAR.DOY
```

`DOY` is the day of the year with three digits. Example: the vertical channel of 15 October 2026 for station `AM.DEMO1.00` is

```
sds/2026/AM/DEMO1/EHZ.D/AM.DEMO1.00.EHZ.D.2026.288
```

Any program that reads SDS (ObsPy, SeisComP, many others) can use the folder directly.

B.2 CSV

The CSV export has a header row, then one row per sample: the UTC time (ISO format with microseconds and a final `Z`) and one column per channel. A missing sample (a gap) is an empty cell. All channels must have the same sampling rate. Example:

```
time_utc,EHZ,EHN,EHE
2026-10-15T03:00:00.000000Z,14997.0,19971.0,-7971.0
2026-10-15T03:00:00.010000Z,15000.0,20025.0,-8007.0
2026-10-15T03:00:00.020000Z,15020.0,20035.0,-8019.0
2026-10-15T03:00:00.030000Z,15036.0,19979.0,-7974.0
2026-10-15T03:00:00.040000Z,14962.0,19984.0,-8020.0
```

B.3 The generated Python script

The **</> Code** button writes a short ObsPy program that reproduces the preview. For one hour of the three channels with the band-pass 1–10 Hz switched on:

```
from obspy import Stream, UTCDateTime
from obspy.clients.filesystem.sds import Client

client = Client('/home/you/.local/share/rs3d/sds')
start = UTCDateTime('2026-10-15T03:00:00.000000Z')
end = UTCDateTime('2026-10-15T04:00:00.000000Z')

st = Stream()
for channel in ('EHZ', 'EHN', 'EHE'):
    st += client.get_waveforms('AM', 'DEMO1', '00', channel, start, end)
st.trim(start, end)

st.detrend("demean")
st.filter("bandpass", freqmin=1.0, freqmax=10.0, corners=4, zerophase=True)

print(st)
```

The first block reads the day files from the data folder; `trim` cuts them to the selection; with **m/s** the program also reads the cached response and calls `remove_response`; with the band-pass it removes the mean and filters. The explanation of each step is in the Theory Manual, Chapters 3 and 4.

B.4 PNG and PDF

The waveform exports are pictures of the selected channels, one above the other, 1200 pixels wide. The helicorder clipboard picture is a 12-by-9-inch figure with three panels (waveform, spectrogram, amplitude spectrum) and a title line naming the station, channel, UTC time range, units and filter.

C Glossary

Amplitude the size of a signal; in RS3D in *counts* or, after conversion, in m/s.

Band-pass filter keeps the frequencies between two limits and removes the rest.

Channel one stream of samples: `EHZ` vertical, `EHN` north–south, `EHE` east–west.

Coda the gradually fading tail of seismic waves after the main arrivals of an earthquake.

Counts the integer numbers produced by the Shake's analogue-to-digital converter; not a physical unit.

Gap time for which no samples were recorded.

Helicorder a plot that wraps a long record into rows, like lines of text.

Hertz (Hz) cycles per second, the unit of frequency.

Link-local address an address of the form `169.254.x.x` that devices give themselves when no router is present.

miniSEED the standard file format for seismic waveform data.

Noise spectrum the typical level of each frequency in a selected time range.

P wave, S wave the first (compressional) and second (shear) arrivals of an earthquake; the S wave arrives later and is usually stronger.

SDS SeisComP data structure: one file per channel per day in a fixed folder layout.

SeedLink the protocol the Shake uses to send data live over the network.

Spectrogram a picture of frequency against time, coloured by power.

SSH a secure login used by RS3D to read the Shake's clock, storage and archive.

UTC coordinated universal time, the time scale used for all seismic data.

References

Raspberry Shake S.A. (2026a). *Raspberry Shake documentation*. URL: https://manual.raspberryshake.org/ (visited on 10/15/2026).
– (2026b). *rsudp: continuous Raspberry Shake UDP data consumer*. URL: https://github.com/raspishake/rsudp (visited on 10/15/2026).